VFS DBVF 26-27 - Mission Model Results

Full-success mission (FM-1 + FM-2 x3 payloads + FM-3 x3 retrievals, end at Home) on one 6S pack

Airframe	Quadcopter, 10 lb unloaded incl. motors, props, battery (R = 1.4); 3 lb water payload
Drag / speed	Cd 1.0 (quadcopter frontal area), 15 m/s governor
Mission	13 takeoffs, 13 landings, 16 transits, 3 hover drops; all transits at 30 ft AGL
Time limit	600 s; throttle chosen per phase for minimum energy inside it
Voltage	6S - every motor curve measured at ~24 V, no pack stacking
Sweep	62 motor/prop combos x 9 batteries = 527 runs
'Usable' means	flies the whole mission, loaded thrust/weight >= 1.8, pack capacity x C covers 4-motor peak current

Recommendation

- Battery: Tattu RLine5 2200 6S 150C (346 g). 1300 mAh finishes too, but with no real reserve.
- Most efficient: Vertiq 60-08 300KV P22x6.6 - 49% of the pack, thrust/weight 3.4, 198 s.
- Best US-brand, published price: KDE4014XF-380 18.5in dual - 60% of the pack, $620 for 4 motors.
- Every usable combo finishes in ~193-203 s: ~397 s of margin on the 600 s limit.

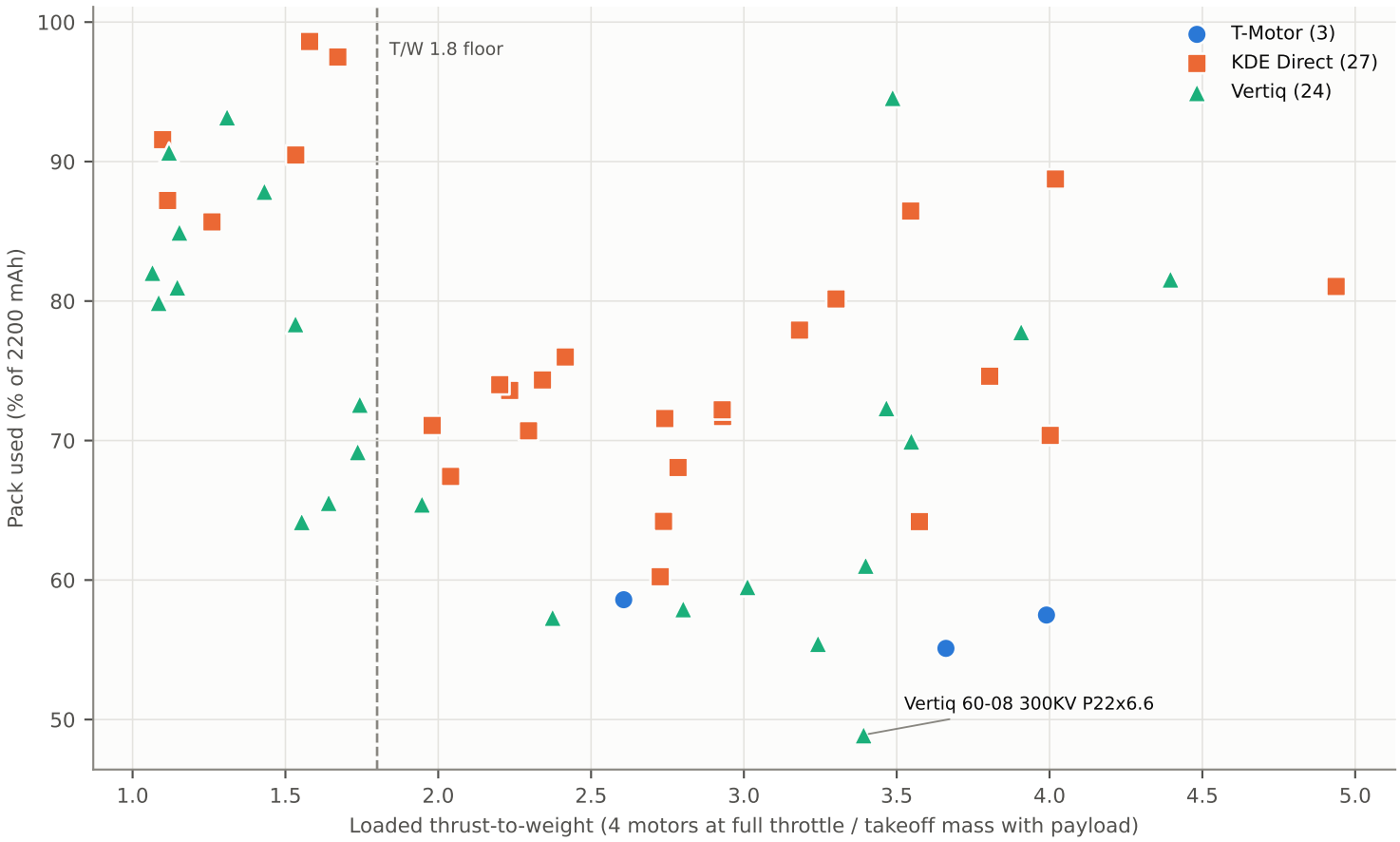
Top 15 motor/prop combos on the RLine5 2200 mAh pack

Motor + prop	Prop in	Mass kg*	T/W loaded	Time s	Energy mAh	Pack used	Peak A	4 motors
Vertiq 60-08 300KV P22x6.6	22	4.54	3.39	198	1076	49%	133	quote
T-Motor MN6007II P21x6.3	21	4.54	3.66	198	1212	55%	155	$727
Vertiq 60-08 300KV P20x6	20	4.54	3.24	199	1220	55%	140	quote
Vertiq 60-08 300KV P17x5.8	17	4.54	2.37	199	1261	57%	90	quote
T-Motor MN6007II P22x6.6	22	4.54	3.99	193	1265	57%	177	$727
Vertiq 60-08 300KV P18x6.1	18	4.54	2.80	199	1275	58%	114	quote
T-Motor MN6007II P18x6.1 (est.)	18	4.54	2.61	197	1289	59%	100	$727
Vertiq 60-08 660KV P20x6	20	4.54	3.01	199	1310	60%	131	quote
KDE4014XF-380 18.5in dual	18.5	4.54	2.73	202	1325	60%	124	$620
Vertiq 60-08 660KV P18x6.1	18	4.54	3.40	200	1343	61%	165	quote
KDE5215XF-330 18.5in dual	18.5	4.54	3.57	199	1412	64%	164	$1040
KDE4213XF-360 18.5in dual	18.5	4.54	2.74	201	1412	64%	125	$680
Vertiq 60-08 300KV P16x5.4	16	4.54	1.95	199	1440	65%	76	quote
KDE4213XF-360 15.5in dual	15.5	4.54	2.04	201	1483	67%	84	$680
KDE5215XF-330 15.5in triple	15.5	4.54	2.78	202	1497	68%	132	$1040

*takeoff mass without payload. Vertiq mass includes its built-in ESC; others exclude the ESC.

Trade space: efficiency vs. thrust margin

Every combo that flies the mission on the 2200 mAh pack. Lower = less battery; right = more thrust headroom.



Best usable combo at each prop size

Prop in	Motor + prop	T/W loaded	Pack used	Peak A
13	Vertiq 60-08 660KV P13x4.4	3.49	95%	251
14	Vertiq 60-08 660KV P14x4.8	4.40	82%	308
15	Vertiq 60-08 660KV P15x5	3.91	78%	248
15.5	KDE4213XF-360 15.5in dual	2.04	67%	84
16	Vertiq 60-08 300KV P16x5.4	1.95	65%	76
17	Vertiq 60-08 300KV P17x5.8	2.37	57%	90
18	Vertiq 60-08 300KV P18x6.1	2.80	58%	114
18.5	KDE4014XF-380 18.5in dual	2.73	60%	124
20	Vertiq 60-08 300KV P20x6	3.24	55%	140
21	T-Motor MN6007II P21x6.3	3.66	55%	155
22	Vertiq 60-08 300KV P22x6.6	3.39	49%	133

Battery sizing for the top picks

Energy left after the full mission. 'no' = can't finish, loaded T/W < 1.8, or C rating too low.

Pack	Pack g	Vertiq 60-08 300KV P22x6.6	T-Motor MN6007II P21x6.3	KDE4014XF-380 18.5in dual
Tattu FunFly 1300 100C	225	no	no	no
Tattu RLine5 2200 150C	346	1124 (51%)	988 (45%)	875 (40%)
Tattu 4500 25C	670	no	no	no
GensAce GTech 5500 60C	828	4424 (80%)	4288 (78%)	4175 (76%)
Tattu GTech 8000 25C	1160	6924 (87%)	6788 (85%)	6675 (83%)
Tattu Plus 10000 25C	1517	8924 (89%)	8788 (88%)	8675 (87%)
Tattu Plus 12000 15C	1670	10924 (91%)	10788 (90%)	10675 (89%)
Tattu GTech 22000 30C	2460	20924 (95%)	20788 (94%)	20675 (94%)
Tattu GTech 30000 25C	3673	> 10 lb	28788 (96%)	28675 (96%)

Reading it: land with >= 20-30% left in practice - the model drains to 0% and does not model voltage sag.

Aircraft is held at 10 lb unloaded: a bigger pack doesn't cut T/W, it eats frame budget.

'> 10 lb' = motors + pack alone exceed 10 lb.

Course as modeled

Aim points from the 26-27 field drawing, feet from Home (H). Pickup spot, A/B/C inset and D center scaled from the picture.

Point	x ft	y ft
H	0	0
S	0	-20
L	-100	0
A	-285	10
B	-285	0
C	-285	-10
D	-317	0

#	Leg	
1	takeoff	
2	cruise	317 ft
3	land	
4	takeoff	
5	cruise	317 ft
6	land	
7	takeoff	
8	cruise	20 ft
9	land	
10	add_payload	
11	takeoff	
12	cruise	287 ft
13	hover	5 s
14	remove_payload	
15	cruise	287 ft
16	land	
17	add_payload	
18	takeoff	
19	cruise	286 ft
20	hover	5 s
21	remove_payload	
22	cruise	286 ft
23	land	
24	add_payload	
25	takeoff	
26	cruise	285 ft
27	hover	5 s
28	remove_payload	
29	cruise	285 ft

#	Leg	
30	land	
31	takeoff	
32	cruise	285 ft
33	land	
34	add_payload	
35	takeoff	
36	cruise	185 ft
37	land	
38	remove_payload	
39	takeoff	
40	cruise	185 ft
41	land	
42	add_payload	
43	takeoff	
44	cruise	185 ft
45	land	
46	remove_payload	
47	takeoff	
48	cruise	185 ft
49	land	
50	add_payload	
51	takeoff	
52	cruise	185 ft
53	land	
54	remove_payload	
55	takeoff	
56	cruise	100 ft
57	land	

All combos on the 2200 mAh pack (1)

Sorted by energy used. Full per-battery results: vfs_dbvf_26-27_results.csv

Motor + prop	Prop in	T/W loaded	Time s	Pack used	Peak A	Usable
Vertiq 60-08 300KV P22x6.6	22	3.39	198	49%	133	yes
T-Motor MN6007II P21x6.3	21	3.66	198	55%	155	yes
Vertiq 60-08 300KV P20x6	20	3.24	199	55%	140	yes
Vertiq 60-08 300KV P17x5.8	17	2.37	199	57%	90	yes
T-Motor MN6007II P22x6.6	22	3.99	193	57%	177	yes
Vertiq 60-08 300KV P18x6.1	18	2.80	199	58%	114	yes
T-Motor MN6007II P18x6.1 (est.)	18	2.61	197	59%	100	yes
Vertiq 60-08 660KV P20x6	20	3.01	199	60%	131	yes
KDE4014XF-380 18.5in dual	18.5	2.73	202	60%	124	yes
Vertiq 60-08 660KV P18x6.1	18	3.40	200	61%	165	yes
Vertiq 40-14 400KV P18x6.1	18	1.55	201	64%	57	fails T/W
KDE5215XF-330 18.5in dual	18.5	3.57	199	64%	164	yes
KDE4213XF-360 18.5in dual	18.5	2.74	201	64%	125	yes
Vertiq 60-08 300KV P16x5.4	16	1.95	199	65%	76	yes
Vertiq 40-14 400KV P17x5.8	17	1.64	201	66%	64	fails T/W
KDE4213XF-360 15.5in dual	15.5	2.04	201	67%	84	yes
KDE5215XF-330 15.5in triple	15.5	2.78	202	68%	132	yes
Vertiq 40-14 400KV P16x5.4	16	1.74	202	69%	73	fails T/W
Vertiq 60-08 660KV P17x5.8	17	3.55	197	70%	191	yes
KDE5215XF-330 18.5in triple	18.5	4.00	199	70%	211	yes
KDE3520XF-400 15.5in dual	15.5	2.30	202	71%	110	yes
KDE4014XF-380 15.5in dual	15.5	1.98	200	71%	84	yes
KDE3520XF-400 18.5in dual	18.5	2.74	203	72%	158	yes
KDE4213XF-360 18.5in triple	18.5	2.93	201	72%	148	yes
KDE4014XF-380 18.5in triple	18.5	2.93	200	72%	148	yes
Vertiq 60-08 660KV P16x5.4	16	3.47	200	72%	197	yes
Vertiq 40-14 400KV P15x5	15	1.74	201	73%	76	fails T/W
KDE4213XF-360 15.5in triple	15.5	2.23	200	74%	102	yes
KDE4014XF-380 15.5in triple	15.5	2.20	200	74%	103	yes
KDE5215XF-330 15.5in dual	15.5	2.34	199	74%	102	yes
KDE4215XF-465 18.5in dual	18.5	3.80	202	75%	250	yes
KDE3520XF-400 15.5in triple	15.5	2.42	201	76%	135	yes
Vertiq 60-08 660KV P15x5	15	3.91	200	78%	248	yes
KDE4215XF-465 15.5in dual	15.5	3.18	199	78%	180	yes
Vertiq 40-14 400KV P14x4.8	14	1.53	202	78%	69	fails T/W
Vertiq 40-06 370KV P15x5	15	1.09	228	80%	40	fails T/W
KDE4215XF-465 15.5in triple	15.5	3.30	203	80%	214	yes
Vertiq 40-06 370KV P14x4.8	14	1.15	215	81%	46	fails T/W
KDE5215XF-435 18.5in dual	18.5	4.94	197	81%	307	yes
Vertiq 60-08 660KV P14x4.8	14	4.40	200	82%	308	yes

All combos on the 2200 mAh pack (2)

Sorted by energy used. Full per-battery results: vfs_dbvf_26-27_results.csv

Motor + prop	Prop in	T/W loaded	Time s	Pack used	Peak A	Usable
Vertiq 40-06 370KV P16x5.4	16	1.07	239	82%	38	fails T/W
Vertiq 40-06 770KV P14x4.8	14	1.15	214	85%	48	fails T/W
KDE3520XF-400 12.5in triple	12.5	1.26	203	86%	54	fails T/W
KDE5215XF-435 15.5in dual	15.5	3.55	197	86%	205	yes
KDE4014XF-380 12.5in triple	12.5	1.11	218	87%	44	fails T/W
Vertiq 40-06 770KV P13x4.4	13	1.43	205	88%	73	fails T/W
KDE5215XF-435 15.5in triple	15.5	4.02	197	89%	251	yes
KDE4215XF-465 12.5in dual	12.5	1.53	200	90%	72	fails T/W
Vertiq 40-14 400KV P13x4.4	13	1.12	223	91%	51	fails T/W
KDE3520XF-400 12.5in dual	12.5	1.10	222	92%	45	fails T/W
Vertiq 40-06 770KV P12x4	12	1.31	206	93%	66	fails T/W
Vertiq 60-08 660KV P13x4.4	13	3.49	199	95%	251	yes
KDE4215XF-465 12.5in triple	12.5	1.67	199	97%	85	fails T/W
KDE2814XF-515 12.5in dual	12.5	1.58	202	99%	88	fails T/W
KDE2814XF-515 12.5in triple	12.5	1.69	-	-	100	can't fly
Vertiq 60-08 660KV P12x4	12	3.10	-	-	222	can't fly
KDE4213XF-360 12.5in triple	12.5	1.02	-	-	41	can't fly
Vertiq 40-06 370KV P17x5.8	17	0.99	-	-	33	can't fly
Vertiq 40-06 370KV P13x4.4	13	1.00	-	-	37	can't fly
Vertiq 40-06 370KV P12x4	12	0.89	-	-	34	can't fly
KDE4213XF-360 12.5in dual	12.5	0.90	-	-	33	can't fly
KDE4014XF-380 12.5in dual	12.5	0.93	-	-	37	can't fly

Data sources & caveats

Motor curves (all measured at ~24 V / 6S):

T-Motor MN6007II KV320 - T-Motor bench sheet, P21x6.3 and P22x6.6.

T-Motor MN6007II + P18x6.1 - ESTIMATED by estimate_prop_curve.py from the 21/22 in data (method checked on KDE data: thrust within 6% at 50-100% throttle; current padded +11%).

KDE Direct - KDE performance sheets (tested at 3,500 ft; sea-level thrust ~10% higher).

Vertiq 40-06 / 40-14 / 60-08 - vertiq.co/thrust-data, 24 V, continuous-rated rows only.

Batteries - manufacturer net weight and genstattu.com list price (Sept 2026).

Props - KDE blade weights; T-Motor P-series weights/prices from ligpower.com.

Caveats:

Energy is drained to 0%; real packs sag under load - keep a 20-30% reserve.

Throttle is never modeled below each sheet's lowest point (40% for T-Motor), so light-load hover current is over- rather than under-estimated.

Landing is a prescribed descent profile, not optimized.

Coaxial losses, wind, and control margin beyond thrust/weight are not modeled.

Vertiq prices are quote-only (shown as 'quote').

Reproduce:

```
python VFS_DBVF_26-27.py --data-dir Motors --battery-file Batteries \
    --vehicle-file vechicle_configs --no-plot --grid-step 1
python make_result_sheet.py
```